Genetics in the Cholestatic Infant: How Often Sequencing Finds the Answer, Rapid Genome Sequencing in the Neonatal Intensive Care Unit, and the Decision About the Kasai Operation.

Picture a five-week-old girl referred because she is still yellow. Her conjugated bilirubin, the fraction that the liver has already processed but cannot excrete into bile, is five milligrams per deciliter. Her stools are pale but not white. Her gamma-glutamyl transferase, the bile duct enzyme usually called GGT, is around three hundred units per liter. The ultrasound shows a small gallbladder. Her pediatrician heard a soft murmur at birth. The surgeons are ready to perform a cholangiogram, an X-ray picture of the bile ducts taken with contrast, later this week, and if the ducts do not fill they will go straight on to a Kasai operation, the procedure in which the blocked ducts at the liver hilum are removed and a loop of bowel is sewn onto the cut surface so that bile can drain. Someone on the team asks whether we should send a genetic test first. Someone else asks whether we can afford to wait for it.

The clinical question of this chapter is when a cholestatic infant should be sequenced, which test should be sent, and what the result is allowed to change. The main idea is simple to state. Sequencing is no longer a late test reserved for the infant in whom everything else has been negative. A rapid genome can now return inside the window in which the Kasai decision is made. That makes genetics useful at the bedside, but it also creates a new temptation, which is to wait for the result. The rule that runs through this chapter is that a genetic result can remove the reason to operate, but it can never supply a reason to delay the cholangiogram.

A note on sources. The framework comes from the 2017 joint guideline on cholestatic jaundice in infants written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, and from the 2024 clinical practice guideline on genetic cholestatic liver disease from the European Association for the Study of the Liver, known as EASL. The yield data come from a Japanese panel study by Togawa and colleagues in 2016, the large American laboratory series of Karpen and colleagues in 2021 and its 2026 extension by Hoskins and colleagues, several Chinese cohorts from Shanghai and elsewhere, and a Turkish neonatal series published in 2025. The intensive care data come from the randomized trials called NSIGHT1 and NSIGHT2, the Australian rapid sequencing programs reported by Lunke and colleagues in 2020 and 2023, the Californian Project Baby Bear, and the head-to-head comparison called GEMINI, published in 2023. The data on Alagille syndrome and the Kasai operation come from the Philadelphia study of Kaye and colleagues in 2010, a 2018 meta-analysis by Fujishiro and colleagues, a Czech cohort from 2015 and a Chinese series from 2025.

Part one. Two clocks run at the same time.

Every cholestatic infant who might have biliary atresia is on a clock. Biliary atresia is a progressive inflammatory and fibrosing destruction of the bile ducts outside the liver, and the Kasai operation works best when it is done early. In the French national cohort reported by Serinet and colleagues in 2009, which included about seven hundred and forty children, the median age at the Kasai operation was sixty days. Survival with the child's own liver was fifty-seven percent at two years, thirty-eight percent at five years and about twenty-eight and a half percent at fifteen years, and it fell steadily with each later fifteen-day band of age at surgery. A 2023 meta-analysis by Hoshino and colleagues found that operating at thirty days of age or younger was associated with better native-liver survival, a relative benefit of about one point two two, and that later surgery roughly doubled the hazard of needing a transplant.

The second clock is the laboratory's. Here the numbers have changed dramatically. A typical commercial cholestasis panel, which reads a fixed list of a few dozen genes, took a median of twenty-one days in the American series of 2021, and three to four weeks in the more recent extension of the same laboratory's data. By contrast, an ultra-rapid exome in the Australian program returned a result in a median of three days. In Project Baby Bear, a rapid genome took on average about three days. In the GEMINI study, the urgent rapid genome took about eighty hours, which is a little over three days, while the urgent targeted panel took about ninety-nine hours. In the earlier NSIGHT1 trial, the median time from enrolment to diagnosis with rapid genome sequencing was thirteen days.

Put the two clocks side by side and the implication is clear. A rapid genome or rapid exome sent on the day of referral can be back before the cholangiogram. A standard cholestasis panel usually cannot. That is the reason this lesson exists, and it is also the reason for the rule. When the result can arrive in time, it should be used. When it cannot, nobody should wait for it.

Part two. Yield is a property of the patient, not of the test.

The single most important fact about diagnostic yield is that it depends on who is tested. With very similar technology, the reported yield in cholestatic infants ranges from about four percent to seventy-one percent, a sixteen-fold difference. The way to understand that range is through pre-test probability, which simply means how likely a genetic cause is before the test is sent.

At the top of the range, the Japanese study by Togawa and colleagues used an eighteen-gene panel in one hundred and nine infants with neonatal or infantile intrahepatic cholestasis. When the clinicians already suspected a genetic cholestasis from the phenotype, seventy-one percent of infants, twenty-two of thirty-one, had a molecular diagnosis. When the cholestasis was unexplained but the infant was otherwise uncomplicated, the yield was twelve and a half percent. And when the cholestasis was unexplained but accompanied by complications such as prematurity, the yield was only four point three percent, two of forty-six. The overall figure, twenty-six percent, is a blend of those three very different groups.

Other cohorts fall along the same gradient. A Taiwanese panel study found a yield of about thirty-two percent overall, rising to sixty-two and a half percent in progressive or syndromic cholestasis of infancy. A 2026 Chinese study of three hundred and twenty-six infants found biliary

atresia in almost half, and among the infants who did not need surgery and were tested after atresia had been excluded, sixty percent carried pathogenic or likely pathogenic variants. A Shanghai cohort of eight hundred and nine children with suspected inherited cholestasis had a yield of about thirty-four percent. A Turkish series of three hundred and seventy-eight neonates had a yield of twenty-eight percent, rising from about eighteen percent before 2010 to about thirty-five and a half percent afterwards as testing improved.

At the bottom of the range are the open-referral laboratory series and the intensive care populations. In the American laboratory series of two thousand one hundred and seventy-one cholestatic patients tested with a sixty-six-gene panel, the yield was twelve percent. When the same laboratory's data were extended to almost eleven thousand samples, the definitive yield fell to six point eight percent, with a further two point two percent of potential diagnoses. About half of those samples came from infants under one year of age. In a Chinese neonatal intensive care unit, one hundred and sixty cholestatic neonates were identified among about five thousand admissions. Almost half had cholestasis associated with parenteral nutrition, which means intravenous feeding, and a clinical exome solved only six point four percent.

The message for the bedside is that the infant tells you how likely the test is to help. A syndromic face, a murmur, a family history, consanguinity, a low GGT, or progressive disease in a term baby all raise the yield. A preterm infant on parenteral nutrition with sepsis and surgery has a cause that is usually visible already, and sequencing should wait until the cholestasis fails to improve once those insults are removed. The Shanghai data add one more useful observation. All seventeen genetic disorders in that cohort presented with onset in the first three months of life, while only five presented after four months. The first three months are where genetic cholestasis lives.

Part three. What sequencing finds in infants depends on where the baby was born.

The genes that sequencing finds differ by population, and a panel should match the spectrum of the population it serves. In the American series of almost eleven thousand panels, the commonest definitive diagnoses were in JAG1, the main Alagille gene, with one hundred and ninety-two cases, followed by SERPINA1, the alpha-1 antitrypsin gene, with one hundred and forty-two, ABCC2 with seventy-two, ABCB11 with forty-five, CFTR, the cystic fibrosis gene, with thirty-eight, POLG, a mitochondrial gene, with thirty-one, and NOTCH2, the second Alagille gene, with twenty-five.

In Japan, among the one hundred and nine infants in the Togawa study, there were twelve infants with Alagille syndrome, five with neonatal Dubin-Johnson syndrome, which is caused by ABCC2, five with citrin deficiency, which is caused by SLC25A13, and six with low-GGT progressive familial intrahepatic cholestasis or its milder relative, benign recurrent intrahepatic cholestasis. In Shanghai, citrin deficiency led the list at about forty-four percent of genetic diagnoses, followed by JAG1 at about twenty-four and a half percent, ABCB11 at eleven percent and ATP8B1 at about six percent. In Turkey, where consanguinity was present in fifty-six percent of the solved cases, the list was dominated by recessive conditions: ATP8B1, ABCB11, ABCB4, DCDC2, DGUOK, KIF12,

USP53, HSD3B7 and a peroxisomal gene. In Korea, six of one hundred and thirty-five infants with neonatal cholestasis, about four and a half percent, had neonatal Dubin-Johnson syndrome, and those infants had higher bilirubin but lower transaminases than infants with other causes.

Two practical lessons follow. First, neonatal Dubin-Johnson syndrome and citrin deficiency must be on any panel used in East Asian infants, and both are conditions in which surgery is never needed. Second, alpha-1 antitrypsin deficiency is one of the commonest genetic diagnoses in North American infants, yet it is diagnosed faster and more cheaply by protein typing. The 2017 guideline places the alpha-1 antitrypsin phenotype, called the Pi type, together with the serum level, in the first tier of the work-up, and it states that a serum level alone is not enough.

Part four. The GGT is the gene filter.

The best single laboratory value for deciding which genes to think about is the GGT. GGT is made by the cells lining the small bile ducts. When the problem is injury or obstruction of the ducts, GGT rises. When the problem is in the pump that exports bile acids from the liver cell, or in the machinery that builds the canalicular membrane, or in the synthesis of the bile acids themselves, the ducts are never injured and the GGT stays low or normal despite severe cholestasis.

The clearest data on how much this matters come from a Shanghai study by Wang and colleagues published in 2016. Two hundred and seven children with chronic cholestasis were sequenced for ATP8B1 and ABCB11, the genes behind the first two types of progressive familial intrahepatic cholestasis. Thirty-nine children, eighteen point eight percent, had disease in one of these genes. The authors then asked what would have happened if testing had been limited to children with a low peak GGT, using a cut-off that depended on age: below seventy units per liter between two and six months, below sixty between seven and twelve months, and below fifty after the first year. All thirty-nine diagnoses fell within that low-GGT group, so restricting testing would have raised the yield to thirty-five point one percent without missing a single case. The ninety-six children above those cut-offs had no diagnosis in either gene, yet they had consumed almost forty-four percent of the sequencing cost.

In practice, a low or normal GGT in a cholestatic infant points to ATP8B1, ABCB11, TJP2, MYO5B and USP53, to the bile acid synthesis defects caused by genes such as HSD3B7 and AKR1D1, and to the syndromes covered in the previous lesson, arthrogryposis, renal dysfunction and cholestasis syndrome and Smith-Lemli-Opitz syndrome. A high GGT points to JAG1 and NOTCH2, ABCB4, CFTR, alpha-1 antitrypsin deficiency, ductal plate and ciliary disorders such as DCDC2, and, of course, to biliary atresia and choledochal malformation. The Turkish neonatal series confirmed the same split.

There is a caveat. The Shanghai data come from a single center, a Chinese genetic spectrum and an era before broad panels. Today the GGT is best used to read a panel, to choose a single gene when a single gene is all that is available, and to recognize when atresia is unlikely. It should not be used to refuse broad testing to an infant with a convincing phenotype.

Part five. The neonatal intensive care unit has already run the trials.

The intensive care literature on rapid sequencing is now large, and it is worth knowing the landmark studies by name. In NSIGHT1, a randomized trial of sixty-five infants, rapid genome sequencing diagnosed thirty-one percent within twenty-eight days, compared with three percent with standard testing. The median time to diagnosis was thirteen days against one hundred and seven days. Management changed in twenty-two percent against nine percent, although that difference was not statistically significant, and the trial stopped early.

NSIGHT2 randomized two hundred and thirteen infants between rapid genome and rapid exome sequencing. The yields were nineteen and twenty percent, essentially identical, and the time to result was about twelve days in both arms. In the sickest infants, ultra-rapid genome sequencing diagnosed forty-six percent with a median of about two days. Testing the trio, meaning the baby and both parents, rather than the baby alone changed the yield by only about one percentage point.

In Australia, Lunke and colleagues reported ultra-rapid exome sequencing in one hundred and eight critically ill children with a median age of twenty-eight days. Fifty-one percent received a diagnosis, the median report time was three days, and management changed in seventy-six percent of those diagnosed. The same group's 2023 Acute Care Genomics program sequenced two hundred and ninety children: genome sequencing diagnosed forty-seven percent, and adding further laboratory techniques raised that to fifty-four percent. The average time to report was under three days. Critical care management changed in seventy-seven percent of diagnosed children, and in sixty percent the result had major implications that included decisions about surgery and transplantation. In California, Project Baby Bear sequenced one hundred and seventy-eight infants on public insurance, diagnosed forty-three percent in about three days, changed management in thirty-one percent and reported net savings for the health system.

The GEMINI study asked the most practical question of all, which is whether a genome is better than a targeted panel. Four hundred hospitalized infants under one year of age had both tests on the same sample. The rapid genome made a diagnosis in forty-nine percent and the targeted panel in twenty-seven percent. The genome found ninety-five diagnoses that the panel missed, while the panel found nine that the genome missed, so neither test was perfect. For routine requests the panel was faster, about one hundred hours against one hundred and forty-seven, but for urgent requests the genome was faster, about eighty hours against ninety-nine. Care changed after the result in nineteen percent of infants. A cost model posted as a preprint, which means it has not been peer reviewed, estimated the upfront cost at about twelve thousand dollars for the genome and about two and a half thousand for the panel, but suggested that genome-first testing was cheaper overall.

Two points of interpretation matter for the hepatologist. First, the yields of thirty to fifty percent come from infants already selected because a genetic disease was suspected. Second, none of these trials reported a cholestasis subgroup or a subgroup with suspected biliary atresia. The liver data are therefore extrapolated. What does exist is a small study of eighteen children with acute liver dysfunction by Bonser and colleagues, in which rapid genome sequencing diagnosed seven, thirty-nine percent, with a median of four days for the diagnostic results. The diagnoses included ornithine transcarbamylase deficiency twice, Wilson disease, the first type of progressive familial intrahepatic cholestasis, a disorder of the gene NBAS, glycogen storage disease type four and a

disorder of the gene LARS, and five of the seven diagnosed children were being assessed or listed for transplantation when the result arrived. The study that pediatric hepatology still needs is a prospective one in which every infant with conjugated hyperbilirubinemia receives a rapid trio genome at referral, with the Kasai decision as the end point. It has not been done.

Part six. The Kasai in the wrong disease.

Why does any of this matter for the Kasai decision? Because the most important genetic mimic of biliary atresia is Alagille syndrome, and Alagille syndrome treated with a Kasai does badly. Alagille syndrome is a dominant multisystem disorder caused mainly by variants in JAG1, with a smaller share in NOTCH2. Its hallmark in the liver is a paucity of the small bile ducts inside the liver, but in early infancy the ducts can look hypoplastic and the cholangiogram can be misread as atresia.

In the Philadelphia study published by Kaye and colleagues in 2010, nineteen patients with Alagille syndrome who had undergone a Kasai operation were compared with thirty-six matched Alagille patients who had not. Transplantation was needed in forty-seven point three percent of the operated patients compared with thirteen point nine percent of controls. Death occurred in thirty-one point six percent compared with two point eight percent. The bilirubin at six to ten weeks of age was similar in the two groups, about nine point six against eight point seven milligrams per deciliter, so the operated infants did not obviously start out sicker. A 2018 meta-analysis by Fujishiro and colleagues pooled six studies and three hundred and ninety-four Alagille patients. Transplantation followed in sixty-three percent of those who had a Kasai against twenty-three percent of those who did not, an odds ratio of about six and a half, and the odds ratio for death was about fifteen.

We must be honest about causation. Infants who went to surgery may have had more severe cholestasis or more hypoplastic ducts, so the operation may be a marker of severe disease rather than its cause. But it is plausible that an unnecessary laparotomy and a hepatic portoenterostomy, which interrupts the normal biliary tree and invites cholangitis, harm a liver that was never obstructed. The authors' advice was to consider Alagille syndrome before surgery in every infant with high-GGT cholestasis.

How often does this happen? In a 2025 Chinese series of fourteen infants with Alagille syndrome, nine, almost two thirds, underwent surgical exploration for suspected atresia, and one had a Kasai before the JAG1 result came back. The same study offers clues. An abnormal gallbladder on ultrasound was seen in about sixty-four percent of the Alagille infants compared with about ninety-six percent of infants with atresia. The median GGT was three hundred and five against five hundred and fifteen units per liter, so the ranges overlapped widely and GGT alone could not separate them. A cardiac anomaly was present in about seventy-nine percent of the Alagille infants against twenty-five percent of those with atresia. In a Czech cohort published in 2015, five of seventy-two patients who had been treated with a Kasai for presumed atresia, about seven percent, turned out to carry JAG1 null variants. Their biopsies had shown ductular proliferation rather than duct paucity, which is a reminder that early histology can mislead. Three were listed for

transplantation around the age of three, two were transplanted and one died. Finally, the 2017 guideline reminds us that hepatobiliary scintigraphy, the nuclear medicine scan that looks for excretion of tracer into the bowel, has a specificity for atresia of only about sixty-eight to seventy-two percent, because non-excretion also occurs with duct paucity, neonatal hepatitis, low birth weight and parenteral nutrition.

Part seven. Genetics runs beside the cholangiogram, not before it.

These data lead to a parallel pathway. On the day of referral, the infant has a fractionated bilirubin, stool color assessment, a GGT and an ultrasound, and the urgent treatable conditions are excluded: sepsis, galactosemia, hypothyroidism and pituitary failure. On the same day, a genetic sample is sent. Where available, that should be a rapid trio genome. Where it is not, a broad cholestasis panel that includes deletion and duplication analysis is the alternative, together with the alpha-1 antitrypsin Pi type and level and a sweat test or CFTR testing. In the next two or three days the team looks actively for Alagille syndrome: an echocardiogram for the murmur, a spine film for butterfly vertebrae, and an eye examination for posterior embryotoxon, a ring at the edge of the cornea. The 2017 guideline is explicit that Alagille features should be sought before the intraoperative cholangiogram, and it notes that pre-operative testing for cystic fibrosis and Alagille syndrome helps the surgeon interpret the cholangiogram. Then the cholangiogram happens on the atresia timetable. A pending result is never a reason to wait.

What can a result change if it arrives first? A pathogenic JAG1 or NOTCH2 variant in an infant whose biliary tree is patent but hypoplastic argues strongly against a Kasai; a cholangiogram is done only if atresia is still genuinely suspected. Biallelic ABCC2 variants mean neonatal Dubin-Johnson syndrome, which needs reassurance and observation, not surgery. Biallelic SLC25A13 variants mean citrin deficiency, treated with a lactose-free formula enriched in medium-chain triglycerides. Biallelic variants in ATP8B1, ABCB11, TJP2 or USP53 place the infant in the low-GGT familial cholestasis spectrum, where the genotype then guides drug therapy with ileal bile acid transporter inhibitors and decisions about surgical biliary diversion. Two variants in CFTR mean cystic fibrosis, in which the cholestasis is usually managed medically, although inspissated bile may occasionally need drainage.

Equally important is what a result must not change. A single heterozygous variant in ABCB4 or ABCB11 is at most a modifier. A variant of uncertain significance is not a diagnosis. A susceptibility allele for biliary atresia, which we will come to next, is research information. None of these should cancel or delay a cholangiogram in an infant with pale stools.

Part eight. There is no biliary atresia gene test.

It is natural to ask whether sequencing can diagnose biliary atresia itself. It cannot. In an exome study of one hundred and one patients with isolated biliary atresia, no gene reached statistical significance. A genome-wide association signal near the gene ADD3 has been replicated, with an odds ratio of about one point seven in a study of three hundred and seventeen cases, but other candidate genes, including GPC1, ARF6 and EFEMP1, did not replicate in that analysis. In biliary atresia with splenic malformation, the syndromic form that travels with laterality defects, variants

in PKD1L1 were found in eight of sixty-seven patients. A preprint that has not been peer reviewed reported ciliary gene variants in about a third of eighty-nine trios with non-syndromic atresia.

None of these findings shifts the probability of atresia enough to alter the work-up. Their clinical value is elsewhere. Findings related to laterality or cilia point to extrahepatic disease in the heart, kidneys and gut that deserves its own screening. And when parents ask about recurrence, the advice that isolated biliary atresia rarely recurs in siblings is not changed by a negative genome.

Part nine. What a genetic report can hide.

Four pitfalls deserve emphasis. The first is copy number. In a 2025 analysis of Alagille syndrome, whole-gene deletions of JAG1 were found in about six and a half percent of probands, and NOTCH2 deletions in about three and a half percent. A test that only reads the sequence will call these infants negative. Always ask whether deletion and duplication analysis was included.

The second is carrier status. In a Taiwanese panel cohort, more than three quarters of cholestatic children carried at least one heterozygous variant in a cholestasis gene. In another comparison, single-gene variants were found in about twenty-five percent of cholestatic children and twenty-six percent of controls. A single variant in ABCB4 or ABCB11 does not explain neonatal cholestasis.

The third is alpha-1 antitrypsin deficiency, which is common, which is not on every panel, and which is best diagnosed by the Pi type and level that belong in the first tier. In the large American series, one hundred and twenty-five of the one hundred and forty-two definitive SERPINA1 diagnoses were the classic ZZ type.

The fourth is the variant of uncertain significance. In the 2021 American series there were three thousand one hundred and seventeen such variants reported against six hundred and sixty-two pathogenic or likely pathogenic ones, and in the 2026 extension more than sixteen thousand uncertain alleles. Parental testing to establish whether two variants lie on different copies of the gene, matching the variant to the biochemistry and tissue findings, and reanalysis over time are the tools for resolving them. The EASL guideline asks for reanalysis of exome or genome data at least every three years.

Two more points belong here. Neonatal Dubin-Johnson syndrome, the ABCC2 disorder, was the third commonest definitive diagnosis in the American series. It presents as conjugated hyperbilirubinemia with near-normal transaminases and resolves, and recognizing it prevents an unnecessary cholangiogram. And a genome may report secondary findings, meaning medically actionable variants unrelated to the reason for testing. The American College of Medical Genetics and Genomics, known as the ACMG, updated its list in 2025, adding among others CYP27A1, a bile acid synthesis gene. Parents should agree in advance what will be returned.

Part ten. What the guidelines say.

On when to test, the guidelines agree. The 2017 NASPGHAN and ESPGHAN guideline places genetics in its second tier, to be decided with a pediatric hepatologist, with a low threshold for gene panels or exome sequencing. The 2024 EASL guideline recommends testing early in the diagnostic

work-up of infants and children with unexplained cholestasis. The verdict is aligned.

On which test, they differ in emphasis. The North American liver guideline mentions panels and exomes, and the ACMG's 2021 guideline recommends exome or genome sequencing as a first- or second-tier test, but for children with congenital anomalies or developmental delay rather than liver disease. EASL recommends a panel or exome rather than single-gene Sanger sequencing and suggests a genome when the exome is negative.

On rapid genome sequencing in intensive care, there is a gap. No liver guideline addresses it. In England, the national genomic test directory offers rapid trio genome sequencing to acutely unwell children with a likely single-gene disorder whose management depends on the result, with approval from clinical genetics.

On alpha-1 antitrypsin, the guidelines are aligned on early testing, and the 2017 guideline is specific about the Pi type and the level. On looking for Alagille syndrome before surgery, they differ: the North American guideline is explicit, the European genetic guideline does not address it. On reanalysis they differ as well: the ACMG asks laboratories to have a policy but sets no interval, while EASL asks for reanalysis at least every three years.

The material gap is the one this chapter is built around. No guideline states that a pending genetic result must not delay the cholangiogram. The North American guideline asks that the atresia work-up be expedited, especially after six weeks of age, but it was written before tests that return in three days were widely available. That temptation is new, and for now the rule has to come from clinical judgment rather than a guideline.

Part eleven. Traps and pearls.

The traps are worth naming. The first is waiting for the panel: the cholangiogram is postponed, the Kasai happens at seventy days, and native-liver survival is lost. The second is accepting a negative sequencing report without deletion analysis, so a JAG1 deletion is missed and an infant with Alagille syndrome receives a Kasai. The third is a report of two variants in ABCB11 accepted without testing the parents, when both variants turn out to sit on the same copy of the gene; the diagnosis and the drug are then wrong. The fourth is a genome for every cholestatic preterm on parenteral nutrition, which yields about six percent and generates anxiety about uncertain variants without benefit. And the last is a normal report read as proof that the disease is not genetic, when the phenotype has moved on and the data have never been re-read.

The numbers worth carrying are these. Yield runs from about four percent in preterm or parenteral nutrition cholestasis to seventy-one percent when the phenotype looks genetic. A low GGT doubled the yield of sequencing for the first two types of familial cholestasis, from about nineteen to thirty-five percent, without missing a case. In North America, JAG1, SERPINA1 and ABCC2 lead the list; in China, citrin deficiency; in Turkey, recessive low-GGT genes. Rapid exome or genome sequencing returns in two to six days, a standard cholestasis panel in about three weeks. In GEMINI, the genome found a diagnosis in forty-nine percent of infants and the panel in twenty-seven. Alagille syndrome after a Kasai carried a transplant rate of forty-seven against

fourteen percent and a death rate of thirty-two against three percent. Nine of fourteen Alagille infants in one recent series were explored surgically. Whole-gene JAG1 deletions make up about six percent of Alagille syndrome. Heterozygous ABCB4 and ABCB11 variants are about as common in controls as in patients. And a result may cancel a Kasai, but it may never delay a cholangiogram.

What to do differently in clinic.

Let us return to the five-week-old girl with pale stools, a small gallbladder, a GGT of three hundred and a murmur. She stays on the atresia timetable. On the day of referral she has her blood drawn for a rapid trio genome, or, where that is not available, a broad cholestasis panel with deletion analysis, together with an alpha-1 antitrypsin Pi type and level and a sweat test. Over the next two days she has an echocardiogram, a spine film and an eye examination. If those show peripheral pulmonary stenosis, a butterfly vertebra and posterior embryotoxon, or if a JAG1 variant comes back before the operation, the surgeons will be told that this is probably Alagille syndrome and that a Kasai is likely to harm her; a cholangiogram is done only if atresia remains a real possibility. If none of that is found, she has her cholangiogram on schedule, whatever the laboratory is still doing.

Other infants follow the same logic. A four-week-old East Asian infant with a high conjugated bilirubin and normal transaminases should make you think of neonatal Dubin-Johnson syndrome and citrin deficiency, with a citrulline level and a panel before anyone considers surgery. An eight-week-old with a low GGT, itch and normal stools is unlikely to have atresia, and the panel should be read for ATP8B1, ABCB11, TJP2 and USP53, with urine bile acids for the synthesis defects. A preterm infant on parenteral nutrition with sepsis should be sequenced only if the cholestasis fails to improve once those causes are removed. A report of one ABCB11 variant plus an uncertain variant calls for parental testing and, if a biopsy is taken, staining for the bile salt export pump, before anyone calls it a diagnosis. And an infant who has already had a Kasai with poor bile flow and whose panel now shows JAG1 should be re-labelled as Alagille syndrome, assessed for treatment with an ileal bile acid transporter inhibitor, and offered family testing.

So, in clinic tomorrow: send genetics early, in parallel, and with deletion analysis. Prefer a rapid genome when the result could change a decision in the coming days. Look for Alagille syndrome actively before every cholangiogram. Use the GGT to read the result. Treat single heterozygous variants and uncertain variants as hypotheses rather than diagnoses. And never let the laboratory clock slow down the surgical one.